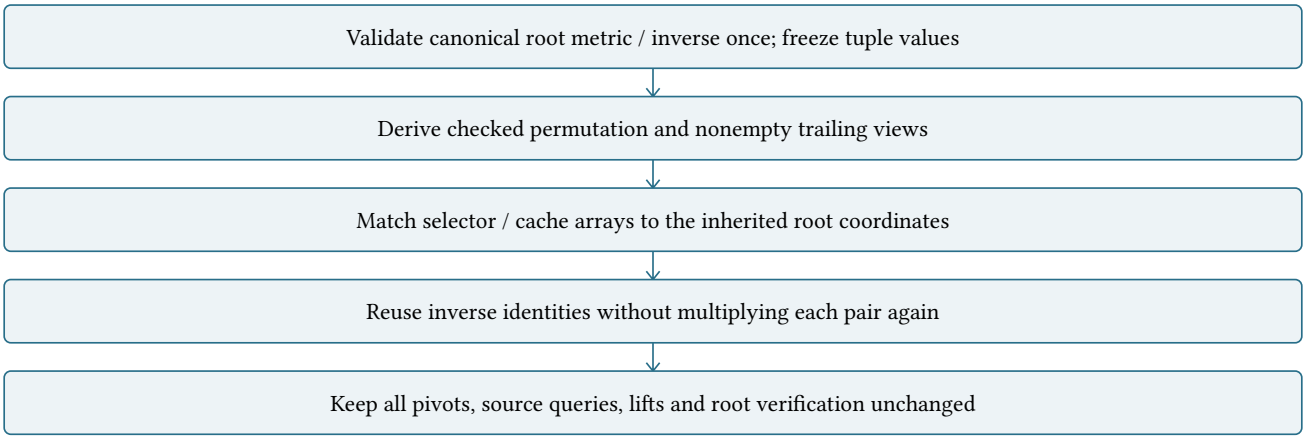
Immutable inverse-metric views

2026-11-02 · Derived validation reuse / static checks · Numerical trials: zero

Invariant. Validate $d_i \cdot di_i = 1$ once at the root. A view stores distinct root indices. Permutations reorder indices; trailing views discard a prefix. Neither changes the field entries, so their inverse identities remain valid. This does not certify $DA = A^T D$; source symmetry remains the compiler/caller's separate obligation.



Entry checks remain. Selector and row-cache entry points require the exact view/root types, same meter, unchanged captured prime, correct dimensions, canonical integer types and matching root-coordinate values. Wrong order or altered values are rejected without repeated field products. Frozen Python objects supply internal provenance, not a public certificate against deliberate reflection or concurrent mutation.

Derived product savings. Let N be root dimension, S the nonzero selector nodes and C the nonterminal cache nodes, with dimensions n_v.

Previous inverse validation	$N + \Sigma(S)n_v + \Sigma(C)n_v$
New inverse validation	N
Removed field products	$\Sigma(S)n_v + \Sigma(C)n_v$
Full scalar descent removed	$N(N+1)-1$
Full pair-only descent removed	$N(N+2)/2-2$, even N
Initially zero source removed	0

Use actual node sums for singular paths. Row reconstruction still uses $2n \cdot$cut products, and all source, transport, pivot and lift arithmetic remains. No inversion or source query is saved by this component.

Host work and storage. Every reuse still checks n coordinate pairs, with native comparisons and index reads. Label/permutation tests, set construction, tuple copying and allocation remain. New counters describe those events; they are not complete CPU or bit-cost measurements. Root metric storage stays 2N field entries. Views share that root and hold index tuples with no parent chain. Existing gathered arrays and frame storage remain; the current view adds index metadata. No runtime or total-heap reduction is measured.

Integration checks. Four declared substitution audits cover the complete driver, selector, row helper and block cache. Complete AST comparisons pass; all Python sources parse. Static inspection finds one root-validation call site and no field multiplication/inversion/source calls in view derivation or coordinate matching. These checks do not execute the new implementation.

Required validation / next. Preserve the full upstream cohort and pivot sequences. Cover zero/scalar/pair and singular terminals, both witness/solution outcomes, nonconstant permuted metrics, wrong values/order/meter/prime, duplicate/out-of-range/bool indices, frozen writes, counters and heap aliases. Compare removed products to the actual node-sum formula independently. Existing cost admission still prevents numerical trials. Next derive safe selector zero-prefix reuse; its query saving is not included here.

Sources. metric_views.py, selection_view.py, rows_view.py, block_cache_view.py, driver_views.py, integration.json and DERIVATION.md. The unchanged exact-field scalar/pair elimination context is Dumas–Pernet2018, arXiv:1802.10453. This representation introduces no new factorization theorem.